

PRABAL KHARE

ROBOTICS SOFTWARE ENGINEER | ROBOTICS ENGINEER | AUTONOMY ENGINEER

Brooklyn, New York | (347) 543 3082 | pk3391@nyu.edu
prabalkhare.com | [linkedin.com/in/prabalk](https://www.linkedin.com/in/prabalk) | github.com/00Prabalk00

EDUCATION

New York University, Tandon School of Engineering

Master of Science in Mechatronics and Robotics | Brooklyn, New York

Expected May 2028

Vellore Institute of Technology Chennai

Bachelor of Technology in Computer Engineering, Robotics and AI Specialization | Chennai, India

July 2026

TECHNICAL SKILLS

Programming: Python, C++, JavaScript and MATLAB

Robotics and autonomy: ROS 2, Nav2, SLAM, tf2, localization, path planning, robot bringup, AMRs, LiDAR, PGV, URDF, RViz, Gazebo, MAVROS and MAVLink

Controls and embedded: CANopen, CiA 402, PID control, sensor fusion, Raspberry Pi, Arduino, Pixhawk, motor drivers and PCB design

Vision and tools: OpenCV, YOLOv8, U-Net, PyTorch, TensorFlow, COLMAP, FEniCSx, PyQt6, Node-RED, Git, Docker, Linux and Web APIs

Spoken languages: English, Hindi, Thai, Marathi and German (basic)

EXPERIENCE

Robotics Software Intern | Next Robotics Lab Co., Ltd.

January 2026 to July 2026

Pathum Thani, Thailand

- Architected a reusable ROS 2 autonomy stack for the 300 kg SMR300 AMR, integrating LiDAR, PGV, Nav2, an AI camera, CANopen and CiA 402 while replacing the company's prior ROS 1 architecture.
- Validated navigation through 150 shelf docking trials and 150 point to point runs, achieving 97% docking success, 2 cm shelf docking error and a 3 cm mean point to point mismatch, down from 5 cm.
- Implemented reflective marker recognition for two and four legged shelves using LiDAR clustering, center checks, automatic repositioning, sector safety controls, safe aborts and alerts; diagnosed failures caused by sunlight reflections at open factory doors.
- Built an operator platform for mapping, map editing, URDF and sensor configuration, Nav2 parameters, mission control, monitoring and deployment validation, plus a Node-RED dashboard evaluated by five operators and two engineers.

Resource Head and Systems Department Member | Dreadnought Robotics

August 2023 to June 2025

Chennai, India | Resource Head, August 2024 to June 2025 | Systems Member, August 2023 to August 2024

- Managed a \$15K+ budget, inventory and procurement decisions for 120 members across four departments, comparing alternatives for cost, availability and system fit.
- Designed electrical systems for reliable operation within a sealed AUV hull and developed ROS, MAVROS and MAVLink control software. The team placed second at TAC Challenge Norway 2024.

Hardware Engineering Intern | Theta Sound

January 2025 to October 2025

Remote

- Produced PCB layouts and simulated prototypes for Qualcomm based smart earphones, evaluating ANC, Bluetooth, PPG, IMU, board footprint, battery life and power; led a KiCad workshop for 50 attendees.

AI Software Development Intern | Drone Entry

June 2025 to August 2025

Thailand

- Built OpenDroneKit, a PyQt6 structural inspection workflow spanning approximately 40,000 metal, brick and other defect images, integrating YOLOv8 detection and U-Net crack segmentation.
- Mapped detected cracks onto COLMAP 3D reconstructions and connected segmentation masks to FEniCSx simulations to estimate crack propagation and flag potential structural severity.

R&D Summer Intern, Robotics and Drones | Drone Academy

May 2025 to June 2025

Thailand

- Designed and fabricated three PCBs for an animatronic head, Raspberry Pi 4 I/O and an RC car kit; ran text to speech and a 2B parameter language model locally on an 8 GB Pi and validated the RC car board for sale.

PROJECTS

Kurat, Autonomous Conversational Robot | ROS 2, PyTorch, YOLO, STT and TTS

2025 to Present

- Designed a three brain perception, cognition and action architecture, integrating a RealSense D435i and Jetson Orin Nano with YOLOv8, Whisper, ORB-SLAM3 and Nav2.

Autonomous VTOL UAV, Search and Rescue | Python, MAVROS, Pixhawk and OpenCV

2023

- Placed in the top 30 of 500 teams at Smart India Hackathon 2023 by building a VTOL search and rescue system with Raspberry Pi, Pixhawk, MAVROS, thermal imaging, TensorFlow Lite and YOLOv5 inference, autonomous scan patterns and a custom battery management system.

PATENTS

Transformation Drone, Autonomous Drone-Rover System

Filed | 202641035669

Thermoregulation System for Portable Water Containers

Filed | Number pending

IMU Calibration and Sensor Fusion Module

Filed | Number pending